

Question	Marking Guidance	Mark	Comments
02.1	1. Calcium ions diffuse into myofibrils from (sarcoplasmic) reticulum; 2. (Calcium ions) cause movement of tropomyosin (on actin); 3. (This movement causes) exposure of the binding sites on the actin; 4. Myosin heads attach to binding sites on actin; 5. Hydrolysis of ATP (on myosin heads) causes myosin heads to bend; 6. (Bending) pulling actin molecules; 7. Attachment of a new ATP molecule to each myosin head causes myosin heads to detach (from actin sites);	5 max	
02.2	1. Releases relatively small amount of energy / little energy lost as heat; 2. Releases energy instantaneously; 3. Phosphorylates other compounds, making them more reactive; 4. Can be rapidly re-synthesised; 5. Is not lost from/does not leave cells;	2 max	1. Key concept is that little danger of thermal death of cells 2. Key concept is that energy is readily available
